SECTION 9 CARRIAGEWAYS: SUB-BASE MATERIAL AND BITUMINOUS MATERIALS

MATERIALS

Aggregates for bituminous materials

9.04 (4) If the *Contractor* finds it necessary to change the source of any aggregate for bituminous materials during the Contract, the *Contractor* shall notify the *Supervisor* in advance and further mix designs trials as specified in this Section will be required.

Recycled bituminous roadbase materials

9.04A (1) Subject to the approval of the *Supervisor*, the *Contractor* may use recycled bituminous roadbase material incorporating Reclaimed Asphalt Pavement (RAP) for carriageway works. This clause contains specific requirements relating to the use of RAP to produce recycled bituminous roadbase material.

(2) Roadbase incorporating RAP shall be designed, manufactured, laid and tested in accordance with the requirements for roadbase using non-recycled materials as specified in Section 9 – "Carriageways: Sub-base Material and Bituminous Materials" of the General Specification for Civil Engineering Works except where modified by this clause or otherwise specified by the *Supervisor*.

(3) Reclaimed Asphalt Pavement (RAP)

(a) RAP shall be obtained from milling or excavation of existing bituminous pavements, and return loads from site and offcuts from bituminous layer joint preparation. Return loads can include bituminous materials rejected from site. RAP shall be crushed and screened as necessary such that the maximum particle size is not greater than the nominal maximum aggregate size of the bituminous roadbase materials being produced and a reasonably well-graded and consistent mixture can be produced.

(b) RAP shall be free of foreign materials such as unbound granular sub-base, broken concrete, or other contaminants.

(c) RAP shall be stored in separate stockpiles prior to use. RAP that has been stockpiled for some time shall be reprocessed, where necessary, to ensure that it is in a free flowing state at the time of use.

(4) The maximum amount of RAP allowable in the bituminous roadbase material shall be 15% by mass. Separate mix designs shall be required for roadbase materials incorporating RAP. When the percentage by mass of RAP incorporated into the roadbase is varied, a separate mix design shall also be required.

Certificate of origin and quality of bitumen

9.06A The *Contractor* shall obtain, keep and supply on request by the *Supervisor*, a Certificate of Origin and a Certificate of Quality for shipment of bitumen pertaining to the *works*.

SUBMISSIONS

Particulars of mixes sub-base material and bituminous material 9.12

(4) The following particulars of wearing course materials shall be submitted to the *Supervisor*:

(a) source and type of aggregates,

(b) grading details in tabular and graphical form,

(c) details of each mixing plant proposed,

(d) certified copies of work sheets for mix designs, which shall include the relative density of the mixed aggregates,

(e) source of bitumen, and

(f) if requested by the *Supervisor*, past test records of same mix produced in the same plant.

(5) The particulars shall be submitted to the *Supervisor* at least 21 days before:

(a) trial areas are constructed, or

(b) the mix is placed in the permanent work if trial areas are not required.

TRIALS

Trial areas 9.17

(1)***S*** Trial areas of each type and layer of bituminous materials shall be constructed to demonstrate that the proposed materials, mixes, methods of production and methods of construction are capable of producing a carriageway which complies with the specified requirements. The *Contractor* shall prepare a trial panel for the proposed pavement materials together with waterproofing system proposed for the viaduct deck. The trial waterproofing and pavement panel shall be laid over a rigid concrete base that is similar in behaviour to the proposed viaduct deck structure. The area of trial panel shall be a minimum of 200m^2; the width of trial panel shall be at least one lane of carriageway and the length of the trial panel shall not be less than 55m. The waterproofing system together with the proposed pavement shall be constructed in accordance with the proposed work procedures and installation method.

Non-compliance: trial areas 9.21

(2)***S*** Unless otherwise permitted by the *Supervisor*, trial areas, or parts of trial areas, which do not comply with the specified requirements for the trial areas shall be removed and rectified by the *Contractor* to the approval of the *Supervisor*.

MIXING OF SUB-BASE MATERIAL AND BITUMINOUS MATERIALS

Mixing bituminous materials 9.29

(4) The *Contractor* shall measure, in the presence of the *Supervisor*, the following temperatures of bituminous materials:

(a) temperature after mixing;

(b) temperature at laying; and

(c) temperature at start of compaction.

LAYING AND COMPACTION OF BITUMINOUS MATERIALS

Laying and compaction of bituminous materials 9.33 (7) The *Contractor* shall ensure that the temperature of bituminous material during laying and compaction shall comply with the requirements of the bridge deck waterproofing manufacturer in order to ensure a robust bond between the wearing course and the waterproofing membrane.

Joints in bituminous materials 9.37 (6)***S*** Unless otherwise permitted by the *Supervisor*, the *Contractor* shall not form joints in friction course by cutting and shall not coat the joints with bituminous emulsion.

(7) For resurfacing of the existing carriageway with 30mm friction course, the existing wearing course shall be scarified by 25mm to receive the 30mm friction course.

PROTECTION OF SURFACES OF SUB-BASE MATERIAL AND BITUMINOUS MATERIALS

Protection of surfaces of sub-base material and bituminous materials 9.38 (4) Constructional Plant or other vehicles travelling from unsurfaced areas shall be kept off friction course.

TOLERANCES

Tolerances: level of carriageway 9.40 (5) If the surface level of the sub-base is not within the tolerances specified in Table 9.9 in GS, the top 75mm shall be scarified, reshaped, with material added or removed as necessary, and then recompacted.

TESTING: BITUMINOUS MATERIALS OTHER THAN BITUMINOUS FRICTION COURSE MATERIAL

Testing: bituminous materials on bridge 9.57A (1) The *Contractor* shall carry out appropriate stripping tests to ensure there is no inherent aggregate stripping deficiency at temperatures expected to be encountered in Hong Kong. The *Contractor* shall submit method statement for the proposed stripping tests to the *Supervisor* for approval before commencement of Works. The method statement shall include, but not limited to, the procedures for carrying out the tests and the guidelines for the interpretation of results. The *Contractor* shall submit all stripping test results to the *Supervisor* for approval.

(2) The *Contractor* shall determine the tensile strength of the bridge wearing course material at various ambient temperatures expected to be encountered in Hong Kong to ensure a value of greater than 0.35 N/mm^2 can be achieved. The *Contractor* shall submit method statement detailing the procedure to determine the tensile strength of the bridge wearing course material to the

Supervisor for approval before commencement of Works.

TESTING: BITUMINOUS MATERIAL CORES

Samples: bituminous material cores 9.62 (6)***S*** Cores shall be taken as soon as practicable but not later than 48 hours after completion of the paving operation, except in periods of high ambient temperatures when the *Supervisor* considers it necessary to delay coring to allow the bituminous material to cool sufficiently. Cores awaiting testing shall be stored in the shade and shall not be stacked on each other or stored adjacent to any heat source to prevent degradation.

TESTING: TEXTURE DEPTH AND PERMEABILITY

Testing: texture depth and permeability 9.66 (6) The test results of texture depth and permeability shall be submitted to the *Supervisor* within 24 hours after carry out the test or before the opening of the tested area to vehicular traffic, whichever is the earlier.

RECTIFICATION

Rectification 9.69 The following gives a guideline for rectification :

(1) Any sub-base or pavement course which is not in accordance with the requirements of this Specification in respect of regularity, surface level tolerance, thickness, material property or air void content is to be rectified by the *Contractor*. The material in the defective areas is to be removed completely down to the next layer of construction or as the *Supervisor* directs. The material may be removed by pneumatic spades, scarifying, cold milling or similar techniques, approved by the *Supervisor*.

(2) Rectification of sub-base and other bituminous pavement courses is recommended as follows:

(a) Sub-base

Where the surface of the sub-base is outside the specified tolerance, the top 75mm shall be scarified, reshaped, with material added or removed as necessary, and then recompacted. Where the material is outside the specified requirements, it shall be completely removed and replaced with material that complies with the requirements of the Specification. The area treated shall not be less than 2m wide and 5m long.

(b) Roadbase, Base Course, Wearing Course and Friction Course

(i) Where the surface of any pavement course is outside the tolerance on the high side, it may be corrected by milling off the excess material provided that the remaining layer thickness is within tolerance. Where the surface is too low, it may generally be made up by either laying the next layer thicker

within reasonable limits or alternatively an appropriate regulating course may be laid. Any rectification is subject to the requirements of Clause 9.40(4).

(ii) Where the material is unsatisfactory, it shall be completely removed and replaced with material which complies with the Specification.

(iii) Areas to be rectified shall be the width laid by the paving machine or one traffic lane wide, whichever is greater and shall be at least 5m long for roadbase and base course or 15m long for wearing course and friction course. Areas of any pavement courses other than friction course requiring rectification should be defined by straight saw cuts not less than 20mm deep. The *Supervisor* may instruct that rectification for friction course be carried out on smaller areas than that required above. In the case of manual laying, the area to rectified shall be at least one traffic lane wide.

(iv) Where the number of irregularities exceeds the limits specified, the area rectified shall be 300m or 75m long as appropriate and the full width of the lanes affected, or such lesser area to be determined by the *Supervisor* to make the surface regularity conform to the required limits.

Excavation and reinstatement in flexible carriageway

9.70 (1) Excavation

Trenches and pits cut in flexible carriageways which are to remain open to traffic following completion of the Contract shall have limits of excavation defined by saw cuts for a minimum depth of 25mm and the remaining excavation shall be carried out in such a manner as to avoid disturbance of the adjacent pavement.

(2) Temporary Reinstatement

Trenches and pits in flexible pavement, which are to remain open to traffic only during the term of the Contract, shall be reinstated in accordance with the Drawings.

Temporary access roads

9.71 (1) Any temporary access roads are to be formed within the site for maintaining public access to existing lots, provision of public access necessary for the completion of the *works* or temporary traffic management of existing carriageway.

(2) The *Contractor* shall ensure that all temporary roads shall be constructed in accordance with the requirements shown on the Drawings.

Quiet equipment and techniques for bituminous road

9.72 The *Contractor* shall follow the guidelines in Appendix M2 on the application of some commonly used quiet equipment and techniques for road pavement works in relation to construction, demolition and re-

pavement works surfacing of bituminous carriageway.

Supply and laying of bituminous surfacing 9.73 Bituminous surfacing shall only be laid by Contractors included in the Specialist Contractors List of Supply of Bituminous Pavement Materials and Construction of Special Bituminous Surfacing.

POLYMER MODIFIED FRICTION COURSE MATERIALS

Polymer modified friction course materials 9.74 (1) Aggregates for Polymer Modified Friction Course Materials

(a) For the purpose of mix design, the combined grading of aggregates for polymer modified friction course materials shall be such that the particle size distribution lies within the limits stated in Table 9.16.

Table 9.16 Design limits for particle size distribution, bitumen and polymer content for polymer modified friction course materials.

Properties		**Type of Materials**
		Polymer Modified Friction Course
Nominal Maximum Aggregate Size (mm)		10 mm
	BS Test Sieve	Percentage by mass passing
Particle Size Distribution	14 mm	100
	10 mm	85 - 100
	6.3 mm	-
	5 mm	20 - 40
	3.35 mm	-
	2.36 mm	5 - 15
	1.18 mm	-
	600 µm	-
	300 µm	-
	150 µm	-
	75 µm	2 - 6
Bitumen content as % of total mass including binder, but excluding polymer	Min.	5.5
	Max.	7.0
Dosage of polymer, % of solid content by weight of binder content		Min. 3

(2) Design of Polymer Modified Friction Course Materials

(a) The materials shall consists of coarse and fine aggregates complying with Clause 9.04, filler complying with Clause 9.05, bitumen complying with Clause 9.06 and polymer complying with Clause 9.74(21). The materials shall have particle size distributions, bitumen contents and polymer dosage within the limits stated in PS Table 9.16.

(b) For polymer modified friction course material, binder drainage tests in accordance with PS Appendix U2 should be carried out on the proposed aggregate/modified binder combination to determine the maximum target binder

content (T_{max}) of the mix. The proposed mix will be acceptable if the maximum target binder content (T_{max}) is equal or greater than the proposed binder content.

(c) The properties of the materials shall be stated in Table 9.17.

Table 9.17 Properties of Designed Bituminous Materials

Properties	**Type of Materials**
	Polymer Modified Friction Course
	Nominal Maximum Aggregate Size (mm)
	10 mm
Marshall Quotient (kN/mm)	-
$= \frac{\text{Marshall Stability}}{\text{Flow}}$	
Air voids in mix as a percentage of total bulk volume	Min. 17.0

(3) Particulars of Filler, Bitumen and Polymer for Polymer Modified Friction Course Materials

(a) The following particulars of the proposed filler, bitumen and polymer for polymer modified friction course materials shall be submitted to the *Supervisor*:

(i) a certificate from the manufacturer for each type of filler showing the manufacture's name, the date and place of manufacture and showing that the filler complies with the requirements stated in the Contract and including results of tests for particle size distribution, and

(ii) a certificate from the manufacturer for bitumen showing the manufacturer's name, the date and place of manufacture and showing that the bitumen complies with the requirements stated in the Contract, including a temperature-viscosity relationship for the bitumen, and including results of tests for :

- Relative Density
- Softening Point
- Penetration
- Ductility
- Retained penetration after thin film over test
- Solubility
- Viscosity

- Loss on heating

(iii) a certificate from the supplier for bituminous materials to confirm the compliance of the polymer with Clause 9.74(21).

(b) The particulars, including certificates, shall be submitted to the *Supervisor* at the time stated in Clause 9.12(3).

(4) Particulars of Mixes for Polymer Modified Friction Course Materials

(a) The following particulars of polymer modified friction course material shall be submitted to the *Supervisor*:

(i) binder content % by weight of total mass excluding polymer,

(ii) type of polymer and its properties,

(iii) dosage % of the polymer by weight of binder content and method of adding and controlling the polymer dosage during production,

(iv) source and type of aggregates,

(v) certified copies of work sheets for mix designs, which shall include the relative density of the mixed aggregates,

(vi) method of blending bitumen and the polymer,

(vii) mixing temperature during production,

(viii) grading details in tabular and graphical form,

(ix) source of bitumen and polymer,

(x) details of each mixing plant proposed,

(xi) ten percent fine values of aggregates, and

(xii) if requested by the *Supervisor*, past test records of the same mix produced in the same plant.

(b) The above particulars shall be submitted to the *Supervisor* at least 14 days before:

(i) trial areas are constructed, or

(ii) the mix is placed in the permanent work if trial areas are not required.

(5) Samples : Trial Areas

One sample of polymer modified friction course materials shall

be provided from each mix used in trial areas. The method of sampling shall be as stated in Table 9.18.

Table 9.18 Sampling and Testing Polymer Modified Friction Course Materials

Type of Material	Properties	Methods of Sampling	Methods of Testing
Polymer Modified Friction Course Material	Particle Size Distribution	Clause 9.74(16)	Clause 9.74(17)
	Polymer Modified Binder Content		
	Texture Depth and Permeability	-	Clause 9.74(19)

(6) Testing : Trial Areas

Each sample of polymer modified friction course material taken as stated in Clause 9.18, shall be tested to determine the properties stated in Table 9.18. The method of testing shall be as stated in Table 9.18.

(7) Approved Mix for Polymer Modified Friction Course Materials

(a) The approved gradation envelope for polymer modified friction course materials shall be the gradation envelope found by applying the tolerances stated in Table 9.19 to the particle size distribution of the approved mix.

Table 9.19 Tolerances for Particle Size Distribution from Approved Mix

BS Test Sieve	Tolerance of Particle Size Distribution in Percentage by Mass of Total Mix Passing BS Test Sieve
	Polymer Modified Friction Course Material
14 mm	0
10 mm	±4
6.3 mm	-
5 mm	±7
3.35 mm	-
2.36 mm	±7
1.18 mm	-
600 µm	-
300 µm	-
150 µm	-
75 µm	±2

(b) The approved polymer modified binder content range for polymer modified friction course material shall be the range formed by applying a tolerance of ±0.5% to the total binder and polymer contents of the approved mix.

(8) Handling and Storage of Polymer Modified Friction Course Materials

Unless otherwise permitted by the *Supervisor*, polymer modified friction course materials shall not be stored in heated surge bins for more than 12 hours and shall not be stored in transport vehicles for more than 3 hours.

(9) Mixing Polymer Modified Friction Course Materials

(a) Aggregates and filler for the materials shall be measured to an accuracy of ±3.0% by mass. The aggregate moisture content after drying shall not exceed 0.4% by mass. The dosage of polymer shall be measured by mass or volume to an accuracy of ±0.5%. If the dosage is measured by volume, it shall be related to the specific gravity of the polymer at the ambient temperature.

(b) The materials shall comply with the temperature requirements as stated in Table 9.20 during and after mixing.

Table 9.20 Temperature Requirements for Polymer Modified Friction Course Materials

Type of Material		Polymer Modified Friction Course
Aggregate temperature at mixing (℃)	Min. Max.	To recommendation of supplier for bituminous materials
Binder temperature at mixing (℃)	Min. Max.	
Polymer modified binder temperature at mixing (℃)	Min. Max.	
Bituminous mixture temperature after mixing (℃)	Min. Max.	
Bituminous mixture temperature at laying (℃)	Min. Max.	
Bituminous mixture temperature at start of compaction (℃)	Min. Max	

(c) The particulars of temperature requirements for polymer modified friction course shall be submitted to the *Supervisor* at the time stated in Clause 9.12(3).

(d) If instructed by the *Supervisor*, the *Contractor* shall measure in the presence of the *Supervisor* the following temperatures :

(i) temperature after mixing;

(ii) temperature at laying; and

(iii) temperature at start of compaction.

(10) Installation of Utilities

Pipes, cables manholes, chamber, gullies and other utilities below carriageways shall be completed and fill material shall be deposited and compacted in trenches before the carriageway is

constructed.

(11) Laying and Compaction of Polymer Modified Friction Course Materials

(a) Polymer modified friction course material shall not be laid when the ambient air temperature is below 10℃. Temperature shall be measured in the shade near to the surface on which laying is to be carried out.

(b) Polymer modified friction course materials shall comply with the temperature requirements as recommended by the supplier for bituminous materials during laying and compaction.

(12) Joints in Polymer Modified Friction Course Materials

No transverse joint is allowed at locations closer than 3 m from any transverse joint of the carriageway or otherwise approved by the *Supervisor*.

(13) Protection of Surface of Polymer Modified Friction Course Materials

Unless otherwise permitted by the *Supervisor*, polymer modified friction course layer shall not be used by Constructional Plant or other vehicles until 6 hours after the material has been compacted.

(14) Tolerances : Level of Carriageway

(a) The levels of the surface of each layer of polymer modified friction course shall be determined on a grid at 10m centres in the longitudinal direction and at 2 m centres in the transverse direction.

(b) The level of the surface of each layer of polymer modified friction course shall be within the tolerances of ± 6 mm.

(c) The difference in level of the surface of polymer modified friction course across joints shall not exceed 3 mm.

(d) The combination of permitted tolerances in levels shall not result in a reduction in the thickness of the polymer modified friction course of more than 5 mm form the specified thickness.

(15) Batch : Polymer Modified Friction Course Materials

A batch of polymer modified friction course is a quantity not exceeding 100t of the materials of the same type and same mix produced at the same mixing plant in one day.

(16) Samples : Polymer Modified Friction Course Materials

(a) One sample of the bituminous materials shall be provided

from each batch of bituminous materials.

(b) The size of each sample shall be at least 15kg.

(c) Sample shall be taken at the mixing plant from the delivery vehicle immediately after loading from the plant or from the surge bin.

(d) Unless otherwise agreed by the *Supervisor*, the method of sampling shall be in accordance with ASTM D 979.

(17) Testing : Polymer Modified Friction Course Materials

(a) Each sample of bituminous materials taken as stated in Clause 9.74(16) shall be tested to determine the particle size distribution and polymer modified binder content.

(b) The method of testing shall be in accordance with the following:-

Particle Size Distribution:	ASTM C136 and ASTM C117
Polymer Modified Binder Content:	PS Appendix U1

(18) Compliance Criteria: Polymer Modified Friction Course Materials

The results of tests on bituminous materials shall comply with the following requirements:

(a) The particular size distribution shall be within the approved gradation envelopes as determined in Clause 9.74(7)(a).

(b) The polymer modified binder content shall be within the approved range as stated in Clause 9.74(7)(b).

(19) Testing : Texture Depth and Permeability of Polymer Modified Friction Course

(a) Testing to determine the texture depth shall be carried out by an independent specialist HOKLAS accredited laboratory approved by the *Supervisor*. The method of testing shall be by the sand patch test in accordance with GS Appendix 10.1.

(b) Testing to determine the permeability shall be carried out an independent specialist HOKLAS accredited laboratory approved by the *Supervisor*. The method of testing shall be in accordance with GS Appendix 9.1.

(c) Rate of tests for texture depth and permeability shall be as specified in GS Table 9.15.

(20) Compliance Criteria : Texture Depth and Permeability of Polymer Modified Friction Course

The results of tests for texture depth and permeability shall comply with the requirements stated in Clauses 9.67 and 9.68 respectively.

(21) Polymer Modified Binder for Polymer Modified Friction Course Materials

(a) The type of polymer used shall be submitted to the *Supervisor* for approval. The properties of the polymer shall also be submitted to the *Supervisor*. The amount of polymer to be used shall give the polymer modified binder the following properties:

Penetration at 25℃ (P)	45 to 60
Softening Point ℃ (SP) (Ring and Ball)	53 to 69

(b) The information required under this sub-clause shall be submitted to the *Supervisor* at least 14 days before trial areas are constructed or the mix is placed in the permanent work if trial areas are not required.

(22) Testing : Polymer Modified Binder Polymer Modified Friction Course Materials.

(a) A batch of polymer modified binder is a quantity not exceeding 10t of polymer modified binder for the production of polymer modified friction course material in one day.

(b) The sample size and method of sampling for polymer modified binder shall be in accordance with ASTM D140 unless otherwise approved by the *Supervisor*.

(c) Each sample of polymer modified binder shall be tested to determine the properties as stated in Table 9.21.

Table 9.21 : Testing polymer modified binder

Property	Method of testing
Penetration at 25℃	ASTM D5
Softening point	BS 2000

(d) A calibration chart of penetration against different percentage of polymer in the polymer modified binder shall be prepared by the supplier for bituminous materials in the presence of the *Supervisor*. From the penetration value obtained in the test carried out in this sub-clause (c), the percentage of polymer in the polymer modified binder is then determined from the calibration chart based on the requirements in Clause 9.74(21)(a).

(e) The testing requirements specified in this Clause are for the pre-blended type polymer modified binder only. For

other types of polymer, preparation of the calibration chart specified in sub-clause (d) is still required.

COLD MILLING WORKS

Equipment and plant 9.75

(1) The equipment or plant used for the texturing, profiling and removal of bituminous pavements shall be a self-propelled and power operated cold milling machine.

(2) The equipment shall be capable of milling existing bituminous surfaces and accurately establishing grades or depths of cut as specified by references from an automatic and independent grade control system.

(3) The grade control system shall be operated independently and controlled at each side of the machine.

(4) The equipment shall be capable of profiling and milling down bituminous materials in varying depths up to 100 mm in a single pass and the minimum single milling width shall have a range from 0.5 m to 2.0 m. Milling depth shall be infinitely variable on each side of the cutting drum to produce variable angles of cutting inclination and road surface profile. The equipment shall be capable of operating within tolerances of ± 4.0 mm. To prevent the development of dust during the milling operation, the machine shall have its own controllable water sprinkler system.

(5) The resultant road surface following texturing, profiling and planning shall show a regular texture suitable and safe for trafficking.

(6) The machine shall preferably be equipped with effective mechanical means for removal of the milled out materials from the planed road surface such as a conveyor belt system, or other effective loading system.

(7) For the texturing or grooving of existing worn or smooth road surfaces, the machine shall be equipped with the specific teeth cutters recommended by the manufacturer to produce the even light texture of grooves at determined spacings along the road surfaces.

(8) The equipment used must be capable of complete sweeping and cleansing of road surfaces after milling. This shall include the use of vacuum type sweepers as well as conventional mechanical brush type equipment.

(9) The *Contractor* is required to provide two milling machines for cold milling works. The first machine shall comply fully with the requirements set out in this Specification. The second may be a smaller machine operating at the lower end of the range of single milling widths specified and shall be a self-driven and power operated cold milling machine. The requirements set out for independent grade control system, milling up to the maximum depth of 100 mm in a single pass and possessing its

own loading and its own controllable water sprinkler system may not be needed for the smaller machine.

(10) The specific equipment to be adopted for milling, the system of loading for milled materials and the system of mechanical sweeping/loading debris shall be approved by the *Supervisor*. The *Contractor* shall furnish full details of all of this equipment he intends to use for this Contract for approval.

(11) All plant and equipment shall, at all times, be maintained in good working condition and the *Contractor* shall provide experienced operators together with maintenance crew, complementary tools and spare parts so that proper upkeep and servicing are regularly maintained and repairs kept to a minimum. This is to ensure that the *works* carried out on public roads are not unduly interrupted for lack of proper maintenance.

Removal of milled materials and cleaning of roads 9.76 (1) The *Contractor* shall supply all supporting equipment, plant, tools, transport, operators, and labour for the speedy and efficient removal of the 'milled out' materials off Site.

(2) The roads and the surrounding areas of the Works Site shall be kept clean and tidy and cleared of all debris and dust during and after the milling operations.

(3) Milled pavements shall be swept and cleaned to the satisfaction of the *Supervisor* with mechanical brush type equipment or, upon instruction from the *Supervisor*, mechanical vacuum road sweeper :

(a) at least once every day until the resurfacing work is completed;

(b) at the time right after cold milling before opening the lane to traffic;

(c) at the time just before commencement of resurfacing work.

Milled-out material 9.77 The milled-out material shall become the property of the *Contractor*, and shall be disposed of by the *Contractor*. The milled-out material shall be removed from Site as soon as possible. In no case shall the time for complete removal exceed 24 hours after the cold milling operation.

Temporary storage of milling plant 9.78 It will not normally be possible to store the milling plant on trafficked roads during non-working hours. The *Contractor* is responsible for the provision of his own parking area for the milling plant, watching and the security of the plant. The prices shall be deemed to have been covered in the Activity Schedule.

Raised manholes 9.79 (1) All manhole covers projecting 50 mm or less above the existing road surface before surfacing/resurfacing work shall be rammed on all sides by the *Contractor* at his own expense with bituminous material to a gradient of not exceeding 10 percent relative to the mean profile of the road in any direction.

(2) Where the projections exceed 50 mm, the manhole covers and frames shall be temporarily removed by the *Contractor* for subsequent use. The *Contractor* shall use temporary covers and displacers such as steel plates to achieve covering of the manhole openings and these shall be strong enough to withstand traffic loading. On completion of laying operation(s) up to the underside of the wearing course of friction course, the *Contractor* shall cut out the roadbase, regulating course and/or base course to remove the temporary covers and displacers and then set the manhole covers and frames in place to the correct line and level. The manhole covers shall then be rammed in accordance with sub-clause (1) of this Clause. Backfilling around the manhole covers and frames shall be completed with a suitable bituminous material complying with GS Tables 9.4 & 9.8. Compacting up to the underside of the wearing course or friction course, shall be performed using hand rammers of other suitable mechanical equipment, in layers not exceeding 50 mm. All prices shall deem to be included in the Activity Schedule.